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Frage 1

Vollständig

Nicht bewertet

By selecting "I confirm", I hereby declare under oath that I will work on this examination on my own without any help or any third-party assistance.

By selecting "I confirm", I understand that noncompliance results in invalidation of the assessment, whereby the invalidated examination will be added to the total number of retakes and noncompliance may result in further legal action.

- ☒ a. I confirm
- ☐ b. I do not confirm

Die richtige Antwort ist: I confirm

Frage 2

Richtig

Erreichte Punkte 1,00 von 1,00

The `__init__` method of a class derived from `torch.nn.Module` ...

- ☐ a. ... is a shortcut for `torch.nn.init`.
- ☐ b. ... initializes all submodules based on the superclass.
- ☐ c. ... specifies how a PyTorch module is applied (the "flow" through the module architecture).
- ☒ d. ... sets up the module architecture. ✓

Die richtige Antwort ist: ... sets up the module architecture.

Frage 3

Richtig

Erreichte Punkte 1,00 von 1,00

Given a sample image `img` (type `PIL.Image`) and a PyTorch transformation pipeline `transforms` (type `torchvision.transforms.Compose`), which of the following is the correct way of applying the transformations on the image?

- ☒ a. `transformed_img = transforms(img)` ✓
- ☐ b. `transformed_img = transforms.apply(img)`
- ☐ c. `transformed_img = img.apply(transforms)`
- ☐ d. `transformed_img = img(transforms)`

Die richtige Antwort ist:

```
transformed_img = transforms(img)
```

Frage 4

Richtig

Erreichte Punkte 1,00 von 1,00

Which of the following are typical data normalization/scaling approaches?

- ☒ a. Scaling to range $[-1, 1]$. ✓
- ☒ b. Scaling to zero (0) mean and unit (1) variance. ✓
- ☐ c. Scaling to range $[0, 100]$.
- ☐ d. Scaling to range $[1, 2]$.

Die richtigen Antworten sind: Scaling to range $[-1, 1]$., Scaling to zero (0) mean and unit (1) variance.

Frage 5

Richtig

Erreichte Punkte 1,00 von 1,00

What can be potential benefits of data augmentation?

- ☐ a. Underfitting can be reduced.
- ☒ b. Overfitting can be reduced. ✓
- ☒ c. The robustness of the model can be increased. ✓
- ☐ d. The robustness of the model can be reduced.

Die richtigen Antworten sind: Overfitting can be reduced., The robustness of the model can be increased.

Frage 6

Richtig

Erreichte Punkte 1,00 von 1,00

What is meant by the term "regularization" in the context of machine learning?

- ☐ a. Input feature normalization.
- ☐ b. Choosing the correct activation function.
- ☐ c. Choosing the correct loss function.
- ☒ d. Techniques to reduce overfitting. ✓

Die richtige Antwort ist: Techniques to reduce overfitting.

Frage 7

Richtig

Erreichte Punkte 1,00 von 1,00

When training a model using some gradient-based iterative method, what does it mean when the training loss increases?

- ☒ a. It means that the model is not learning properly. ✓
- ☐ b. It means that the model outputs are getting closer to the true target values.
- ☐ c. It means that the model is overfitting.
- ☒ d. It means that the model outputs are getting farther away from the true target values. ✓

Die richtigen Antworten sind: It means that the model outputs are getting farther away from the true target values., It means that the model is not learning properly.

Frage 8

Richtig

Erreichte Punkte 1,00 von 1,00

What is a hyperparameter in the setting of machine learning?

- ☐ a. It represents an internal model parameter that is adjusted during training.
- ☒ b. It represents a user-specified parameter that typically influences the model and/or training procedure. ✓
- ☐ c. It represents the process of training and evaluating an algorithm.
- ☐ d. It represents a machine learning model class.

Die richtige Antwort ist: It represents a user-specified parameter that typically influences the model and/or training procedure.

Frage 9

Richtig

Erreichte Punkte 1,00 von 1,00

Which of the following is the correct order of performing one weight update in PyTorch (you can assume that `optimizer` is a valid PyTorch optimizer object)?

- ☐ a. 1. Compute `loss`
2. Update weights `optimizer.step()`
3. Compute gradients `loss.backward()`
4. Reset gradients `optimizer.zero_grad()`
- ☒ b. 1. Compute `loss` ✓
2. Compute gradients `loss.backward()`
3. Update weights `optimizer.step()`
4. Reset gradients `optimizer.zero_grad()`
- ☐ c. 1. Compute `loss`
2. Compute gradients `loss.backward()`
3. Reset gradients `optimizer.zero_grad()`
4. Update weights `optimizer.step()`
- ☐ d. 1. Reset gradients `optimizer.zero_grad()`
2. Compute `loss`
3. Update weights `optimizer.step()`
4. Compute gradients `loss.backward()`

Die richtige Antwort ist: 1. Compute `loss`
2. Compute gradients `loss.backward()`
3. Update weights `optimizer.step()`
4. Reset gradients `optimizer.zero_grad()`

Frage 10

Teilweise richtig

Erreichte Punkte 0,67 von 1,00

Assume that you have one of many arbitrary 8 bit grayscale images `img` (NumPy array) that you want to globally normalize (=normalization on complete data set) to the range [0, 1]. Which of the following code snippets achieve this normalization?

- ☐ a. `(img - img.mean()) / img.std()`
- ☒ b. `(img - img.min()) / (img.max() - img.min())` ✗
- ☐ c. `img * 0 + (1 * img)`
- ☒ d. `img / 255` ✓

Die richtige Antwort ist:

`img / 255`

Frage 11

Richtig

Erreichte Punkte 1,00 von 1,00

In a study, people are asked how often they went to the cinema during last year. Which type of data is this?

- ☐ a. ordinal data
- ☐ b. continuous numerical data
- ☒ c. discrete numerical data ✓
- ☐ d. categorical data

Die richtige Antwort ist: discrete numerical data

Frage 12

Richtig

Erreichte Punkte 1,00 von 1,00

Which of the following statements are correct regarding full-batch learning?

- ☐ a. The batch size is a hyperparameter.
- ☒ b. It leads to smooth gradients. ✓
- ☒ c. All training samples are used for one update. ✓
- ☐ d. It is the fastest among the three main learning procedures (full-batch, online, minibatch).

Die richtigen Antworten sind: All training samples are used for one update., It leads to smooth gradients.

Frage 13

Teilweise richtig

Erreichte Punkte 0,67 von 1,00

Which of the following statements are correct regarding training, validation and test sets?

- ☒ a. Training, validation and test sets enable the detection of overfitting. ✓
- ☒ b. Training, validation and test sets enable good estimates of the model generalization performance. ✓
- ☐ c. Training, validation and test sets enable Empirical Risk Minimization (ERM).
- ☐ d. Training, validation and test sets enable the detection of underfitting.

Die richtigen Antworten sind: Training, validation and test sets enable good estimates of the model generalization performance., Training, validation and test sets enable the detection of underfitting., Training, validation and test sets enable the detection of overfitting.

Frage 14

Teilweise richtig

Erreichte Punkte 0,67 von 1,00

Assume that `MyModule` is a class properly derived from the `torch.nn.Module` class and `tensor_a` and `tensor_b` are PyTorch tensors. What does the following code do?

```
my_module = MyModule()  
my_module(tensor_a, tensor_b)
```

- ☒ a. It creates an instance of `MyModule`. Then it applies the `forward` method of `my_module` with arguments `tensor_a` and `tensor_b`. ✓
- ☒ b. It raises an exception, since only a single argument can be provided when calling `my_module`. ✗
- ☐ c. It raises an exception, since one cannot create instances of `MyModule`.
- ☐ d. It creates an instance of `MyModule`. Then it applies the `forward` method of `my_module` with argument `tensor_b`. `tensor_a` is ignored.

Die richtige Antwort ist: It creates an instance of `MyModule`. Then it applies the `forward` method of `my_module` with arguments `tensor_a` and `tensor_b`.

Frage 15

Richtig

Erreichte Punkte 1,00 von 1,00

What does independently and identically distributed (i.i.d.) mean?

- ☐ a. Each sample has the same probability distribution as the others and all are mutually dependent.
- ☒ b. Each sample has the same probability distribution as the others and all are mutually independent. ✓
- ☐ c. Each sample has a different probability distribution as the others and all are mutually independent.
- ☐ d. Each sample has a different probability distribution as the others and all are mutually dependent.

Die richtige Antwort ist: Each sample has the same probability distribution as the others and all are mutually independent.

Frage 16

Richtig

Erreichte Punkte 1,00 von 1,00

Which of the following statements are correct regarding the following hash function:

```
def my_hash(x):  
    return int(x)
```

- ☐ a. The hash function will work on any input.
- ☐ b. There cannot be any hash collisions.
- ☒ c. Different inputs could lead to the same hash value. ✓
- ☒ d. The computation of the hash value is fast. ✓

Die richtigen Antworten sind: The computation of the hash value is fast., Different inputs could lead to the same hash value.

Frage 17

Richtig

Erreichte Punkte 1,00 von 1,00

What is the purpose of a loss function?

- ☒ a. To compute the difference between the model output and the actual targets (ground truth). ✓
- ☐ b. To change the model parameters.
- ☐ c. To compute the gradients for gradient-based iterative methods.
- ☐ d. To get the predictions of a model.

Die richtige Antwort ist: To compute the difference between the model output and the actual targets (ground truth).

Frage 18

Richtig

Erreichte Punkte 1,00 von 1,00

What is git?

- ☐ a. A neural network architecture.
- ☐ b. A Python package for numerical computations.
- ☐ c. A platform to host programming projects.
- ☒ d. A version control system (VCS). ✓

Die richtige Antwort ist: A version control system (VCS).

Frage 19

Richtig

Erreichte Punkte 1,00 von 1,00

Which of the following are common loss functions for classification tasks?

- ☐ a. Mean-squared error
- ☒ b. Cross entropy ✓
- ☐ c. Sigmoid
- ☐ d. Stochastic gradient descent

Die richtige Antwort ist: Cross entropy

Frage 20

Richtig

Erreichte Punkte 1,00 von 1,00

Why is the following code problematic (you can assume correct inputs, shapes and arguments)?

```
import torch

class MyModule(torch.nn.Module):
    def __init__(self):
        super().__init__()
        self.conv = torch.nn.Conv2d(...)

    def forward(self, x):
        out1 = self.conv(x)
        ... # process "out1" so it can be an input to "linear"
        linear = torch.nn.Linear(...)
        out2 = linear(out1)
        return out2
```

- ☒ a. The submodule `torch.nn.Linear` is created inside the `forward` method rather than the `__init__` method. This means that the submodule is repeatedly created in every `forward` call, including its random parameter initialization. ✓
- ☐ b. `torch.nn.Linear` must be invoked before passing the data to the convolutional layer.
- ☐ c. `torch.nn.Linear` is assigned to a local identifier but `torch.nn.Conv2d` is not. Both should be defined to locals or the code will suffer from reproducibility.
- ☐ d. `torch.nn.Conv2d` can never be combined with `torch.nn.Linear`.

Die richtige Antwort ist: The submodule `torch.nn.Linear` is created inside the `forward` method rather than the `__init__` method. This means that the submodule is repeatedly created in every `forward` call, including its random parameter initialization.

Frage 21

Richtig

Erreichte Punkte 1,00 von 1,00

The special method `__getitem__` in a `torch.utils.data.Dataset` derived class should ...

- ☐ a. ... return minibatched samples.
- ☐ b. ... return all samples.
- ☒ c. ... return one sample. ✓
- ☐ d. ... return the data set.

Die richtige Antwort ist: ... return one sample.

Frage 22

Richtig

Erreichte Punkte 1,00 von 1,00

Consider the following `Dataset` implementation:

```
import numpy as np
from torch.utils.data import Dataset

class MyDataset(Dataset):
    def __getitem__(self, index):
        rng = np.random.default_rng(seed=index)
        return index, rng.random()
```

Which of the following statements are correct?

- ☐ a. The implementation does not work because the mandatory `__len__` method is missing.
- ☒ b. An instance of this class would produce infinitely many samples. ✓
- ☒ c. A sample returned by the `__getitem__` method always contains the index and a randomly generated number. ✓
- ☐ d. In a multi-processed data loading setting, the returned samples might not be reproducible.

Die richtigen Antworten sind: An instance of this class would produce infinitely many samples., A sample returned by the `__getitem__` method always contains the index and a randomly generated number.

Frage 23

Richtig

Erreichte Punkte 1,00 von 1,00

What is the output of the following code?

```
import torch

class MyModule(torch.nn.Module):
    def __init__(self, a):
        self.a = a

    def forward(self, x):
        output = x + self.a
        return output

my_module = MyModule(2.0)
c = my_module(5.0)
print(c)
```

- ☐ a. It raises an exception because the `forward` method is not called.
- ☒ b. It raises an exception because `super().__init__()` is missing in the `__init__` method. ✓
- ☐ c. It raises an exception at `self.a = a` because only PyTorch trainable parameters and (sub)modules can be used as attributes.
- ☐ d. 7.0

Die richtige Antwort ist: It raises an exception because `super().__init__()` is missing in the `__init__` method.

Frage 24

Richtig

Erreichte Punkte 1,00 von 1,00

Which of the following statements are true regarding data augmentation?

- ☒ a. Modifications in data augmentation can often be applied on-the-fly. ✓
- ☒ b. Data augmentation can have a negative impact on the model performance. ✓
- ☒ c. Modifications in data augmentation heavily depend on the data and task. ✓
- ☒ d. Data augmentation is about creating "new" artificial samples by modifying existing samples. ✓

Die richtigen Antworten sind: Data augmentation is about creating "new" artificial samples by modifying existing samples., Modifications in data augmentation can often be applied on-the-fly., Data augmentation can have a negative impact on the model performance., Modifications in data augmentation heavily depend on the data and task.

Frage 25

Richtig

Erreichte Punkte 1,00 von 1,00

What is the output of the following code?

```
import torch

def function(x: torch.Tensor):
    return x if x.min() < 0 else x * -1

scripted_function = torch.jit.script(function)
traced_function = torch.jit.trace(function, example_inputs=torch.tensor([1, 2, 3]))

actual_input = torch.tensor([-4, -5, -6])
print("s:", scripted_function(actual_input).tolist())
print("t:", traced_function(actual_input).tolist())
```

- ☐ a. s: [-4, -5, -6]
t: [-1, -2, -3]
- ☐ b. s: [4, 5, 6]
t: [4, 5, 6]
- ☐ c. s: [4, 5, 6]
t: [1, 2, 3]
- ☐ d. s: [4, 5, 6]
t: [-1, -2, -3]
- ☐ e. s: [-4, -5, -6]
t: [1, 2, 3]
- ☒ f. s: [-4, -5, -6] ✓
t: [4, 5, 6]
- ☐ g. s: [-4, -5, -6]
t: [-4, -5, -6]
- ☐ h. s: [4, 5, 6]
t: [-4, -5, -6]

Die richtige Antwort ist:

```
s: [-4, -5, -6]
t: [4, 5, 6]
```

Frage 26

Richtig

Erreichte Punkte 1,00 von 1,00

Assume that you have a classification task where you want to classify images into either "dog" or "cat". Each image only contains a single dog or a single cat on some arbitrary background. Which of the following data augmentation techniques are useful?

- ☒ a. Applying input dropout ✓
- ☐ b. Zooming into the background
- ☒ c. Adding slight noise ✓
- ☒ d. Flipping horizontally ✓

Die richtigen Antworten sind: Flipping horizontally, Applying input dropout, Adding slight noise

Frage 27

Richtig

Erreichte Punkte 1,00 von 1,00

Ordinal data can be described as:

- ☐ a. Quantitative data with mathematical meaning but without a natural ordering.
- ☐ b. Qualitative data without mathematical meaning.
- ☒ c. Qualitative data without mathematical meaning but with a natural ordering. ✓
- ☐ d. Quantitative data with mathematical meaning and a natural ordering.

Die richtige Antwort ist: Qualitative data without mathematical meaning but with a natural ordering.

Frage 28

Richtig

Erreichte Punkte 1,00 von 1,00

Assume you have the following categorical data in exactly this fixed order:

computer, system, software, hardware

You are asked to create a one-hot encoding. Which of the following is the correct one-hot encoded representation of **hardware**?

- ☐ a. (1, 0, 0, 0)
- ☐ b. (0, 0, 0, 0)
- ☐ c. (1, 1, 1, 1)
- ☐ d. (1, 1, 1, 0)
- ☐ e. (0, 1, 1, 1)
- ☒ f. (0, 0, 0, 1) ✓

Die richtige Antwort ist: (0, 0, 0, 1)

Frage 29

Richtig

Erreichte Punkte 1,00 von 1,00

Which of the following statements are correct regarding monitoring a model during training?

- ☒ a. Monitoring might aid in determining over- or underfitting. ✓
- ☐ b. Monitoring might decrease the model performance.
- ☒ c. Monitoring might help in finding issues during training. ✓
- ☐ d. Monitoring might increase the model performance.

Die richtigen Antworten sind: Monitoring might help in finding issues during training., Monitoring might aid in determining over- or underfitting.

Frage 30

Richtig

Erreichte Punkte 1,00 von 1,00

`torch.utils.data.Subset` can be used to ...

- ☒ a. ... generate a Dataset from a subset of the original Dataset according to a list of indices. ✓
- ☐ b. ... find a good subset of the input features, e.g., for dimensionality reduction.
- ☐ c. ... distribute the sampling process over multiple sub-processes.
- ☐ d. ... derive a custom class that creates minibatches from samples.

Die richtige Antwort ist: ... generate a Dataset from a subset of the original Dataset according to a list of indices.

Frage 31

Richtig

Erreichte Punkte 1,00 von 1,00

In a supervised setting, how can one determine the performance of some machine learning model?

- ☐ a. By averaging the gradients of the model and comparing it to a specified threshold.
- ☐ b. By comparing the model input and the model prediction with each other.
- ☒ c. By using a loss function to compute the deviation between the model prediction and the true target. ✓
- ☐ d. By running a performance profiler on the model.

Die richtige Antwort ist: By using a loss function to compute the deviation between the model prediction and the true target.

Frage 32

Falsch

Erreichte Punkte 0,00 von 1,00

What is the shape of the output tensor `result` when running the following code?

```
import torch

class MyModule(torch.nn.Module):
    def __init__(self):
        super().__init__()
        self.linear = torch.nn.Linear(in_features=4, out_features=6)

    def forward(self, x):
        out = self.linear(x)
        return torch.sigmoid(out)

inp = torch.rand(size=(12, 4))
my_module = MyModule()
result = my_module(inp)
```

- ☐ a. (12, 4)
- ☐ b. (4, 6)
- ☒ c. (12,) ✖
- ☐ d. (12, 6)
- ☐ e. (4,)
- ☐ f. (12, 4, 6)
- ☐ g. (6,)

Die richtige Antwort ist:

(12, 6)

Frage 33

Richtig

Erreichte Punkte 1,00 von 1,00

Which statements regarding PyTorch's `torch.nn.Module` are correct?

- ☒ a. It can contain and utilize other modules. ✔
- ☒ b. It supports the automatic registration of trainable parameters. ✔
- ☒ c. It is the base class for all neural network modules. ✔
- ☒ d. All custom networks/layers/modules should be derived from this class. ✔

Die richtigen Antworten sind: It is the base class for all neural network modules., All custom networks/layers/modules should be derived from this class., It supports the automatic registration of trainable parameters., It can contain and utilize other modules.

Frage 34

Richtig

Erreichte Punkte 1,00 von 1,00

The `forward` method of a class derived from `torch.nn.Module` ...

- ☐ a. ... pushes a PyTorch module to a given device.
- ☐ b. ... adds a new forward layer to the network architecture of a module.
- ☐ c. ... sets up the module architecture.
- ☒ d. ... specifies how a PyTorch module is applied (the "flow" through the module architecture). ✓

Die richtige Antwort ist: ... specifies how a PyTorch module is applied (the "flow" through the module architecture).

Frage 35

Richtig

Erreichte Punkte 1,00 von 1,00

What is the output of the following code?

```
import torch
from torch.utils.data import Dataset, DataLoader

class MyDataset(Dataset):
    def __getitem__(self, index):
        return torch.tensor([1, 2, 3, 4])

    def __len__(self):
        return 12

dataset = MyDataset()
loader = DataLoader(dataset, batch_size=4)

for i, x in enumerate(loader):
    print(f"{i}: {tuple(x.shape)}")
```

- ☐ a. 0: (36,)
- ☐ b. 0: (12, 4, 4)
- ☒ c. 0: (4, 4) ✓
1: (4, 4)
2: (4, 4)
- ☐ d. 0: (4, 12, 4)
- ☐ e. 0: (4,)
1: (4,)
2: (4,)
- ☐ f. 0: (12, 4)
1: (12, 4)

Die richtige Antwort ist:

```
0: (4, 4)
1: (4, 4)
2: (4, 4)
```

Frage 36

Richtig

Erreichte Punkte 1,00 von 1,00

What does `collate_fn` (collate function) do in terms of `torch.utils.data.DataLoader`?

- ☐ a. It specifies the size of the subsets of the dataset.
- ☐ b. It specifies how a sample is loaded from the disk.
- ☐ c. It applies data augmentation methods to the samples.
- ☒ d. It specifies how the samples are combined into minibatches. ✓

Die richtige Antwort ist: It specifies how the samples are combined into minibatches.

Frage 37

Richtig

Erreichte Punkte 1,00 von 1,00

If you start a new machine learning project, which of the following is one of the first steps that you should take?

- ☐ a. Renting cloud-based hardware.
- ☐ b. Implementing the code for training and testing a neural network model.
- ☐ c. Creating training, evaluation and test datasets.
- ☒ d. Looking into the current state of research. ✓

Die richtige Antwort ist: Looking into the current state of research.

Frage 38

Richtig

Erreichte Punkte 1,00 von 1,00

Training a neural network for one epoch will ...

- ☐ a. ... perform training for a fixed number of seconds.
- ☐ b. ... perform training until the model overfits on all training samples.
- ☒ c. ... perform one training iteration over all training samples. ✓
- ☐ d. ... perform one weight update.

Die richtige Antwort ist: ... perform one training iteration over all training samples.

Frage 39

Richtig

Erreichte Punkte 1,00 von 1,00

What is a hash collision?

- ☒ a. Different inputs result in the same hash code. ✓
- ☐ b. Equal inputs result in the same hash code.
- ☐ c. Different inputs result in different hash codes.
- ☐ d. Equal inputs result in different hash codes.

Die richtige Antwort ist: Different inputs result in the same hash code.

Frage 40

Richtig

Erreichte Punkte 1,00 von 1,00

Assume you have the following binary confusion matrix:

		Predicted	
		positive	negative
Actual	positive	0	60
	negative	0	40

Which of the following statements are correct?

- ☒ a. The total number of samples is 100. ✓
- ☒ b. The accuracy is 40%. ✓
- ☒ c. The model is more likely to predict the negative class. ✓
- ☐ d. The model did not make any mistakes with respect to the actual positive samples.

Die richtigen Antworten sind: The accuracy is 40%, The total number of samples is 100., The model is more likely to predict the negative class.

Frage 41

Richtig

Erreichte Punkte 1,00 von 1,00

Assume you apply the following PyTorch image transformation to some example RGB image:

```
transforms.Compose([
    transforms.RandomHorizontalFlip(0.5),
    transforms.RandomVerticalFlip(0.5),
    transforms.RandomRotation(degrees=180),
    transforms.ToTensor()
])
```

Which of the following statements are correct?

- ☒ a. The resulting image might not be flipped (neither horizontally nor vertically) or rotated at all. ✓
- ☐ b. The resulting image will be rotated by 180 degrees.
- ☐ c. The resulting image will be vertically flipped.
- ☐ d. The resulting image will be horizontally flipped.
- ☒ e. The resulting image will be a PyTorch tensor. ✓

Die richtigen Antworten sind: The resulting image will be a PyTorch tensor., The resulting image might not be flipped (neither horizontally nor vertically) or rotated at all.